Introduction to Java

JAVA PROGRAMMING

Objectives

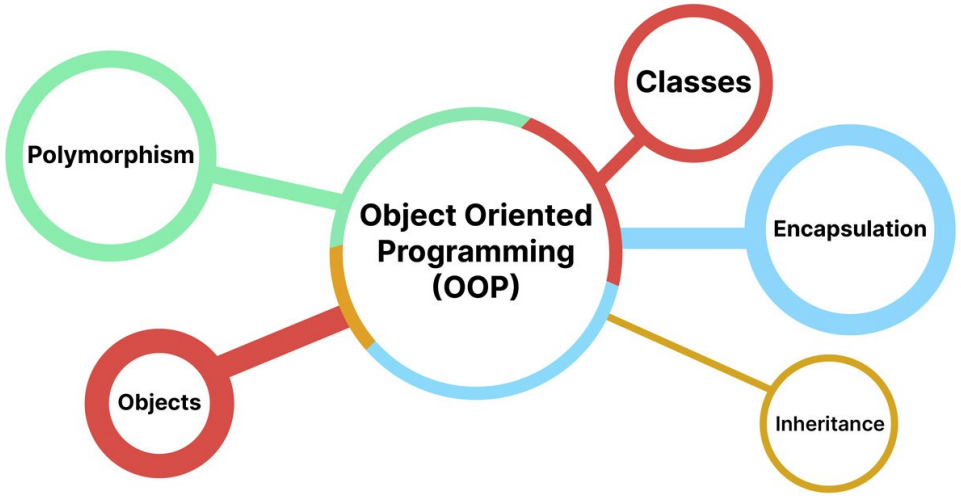
- Become familiar with the Java language.
- Explore Java language editions.
- Become familiar with why Java language is important.
- Explore Java as an independent platform
- Become familiar with Java platform components (Java compiler, Java development kit (JDK),
  Java runtime environment (JRE), Java virtual machine (JVM) and Java just in time compiler(JIT).
- Explore the relationship between JDK,JRE,JVM and JIT.
- Explore how Java program is processed

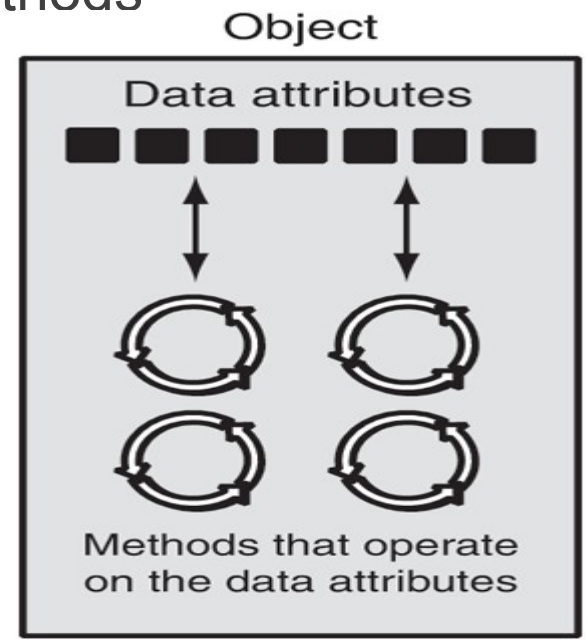
Procedural Programming

- Procedural programming: writing programs made of functions that perform specific tasks
  - Procedures typically operate on data items that are separate from the procedures
  - Data items commonly passed from one procedure to another
  - Focus: to create procedures that operate on the program's data

Object-Oriented Programming

- Object-oriented programming: focused on creating objects
- Object: entity that contains data and procedures
  - Data is known as data attributes and procedures are known as methods
    - Methods perform operations on the data attributes



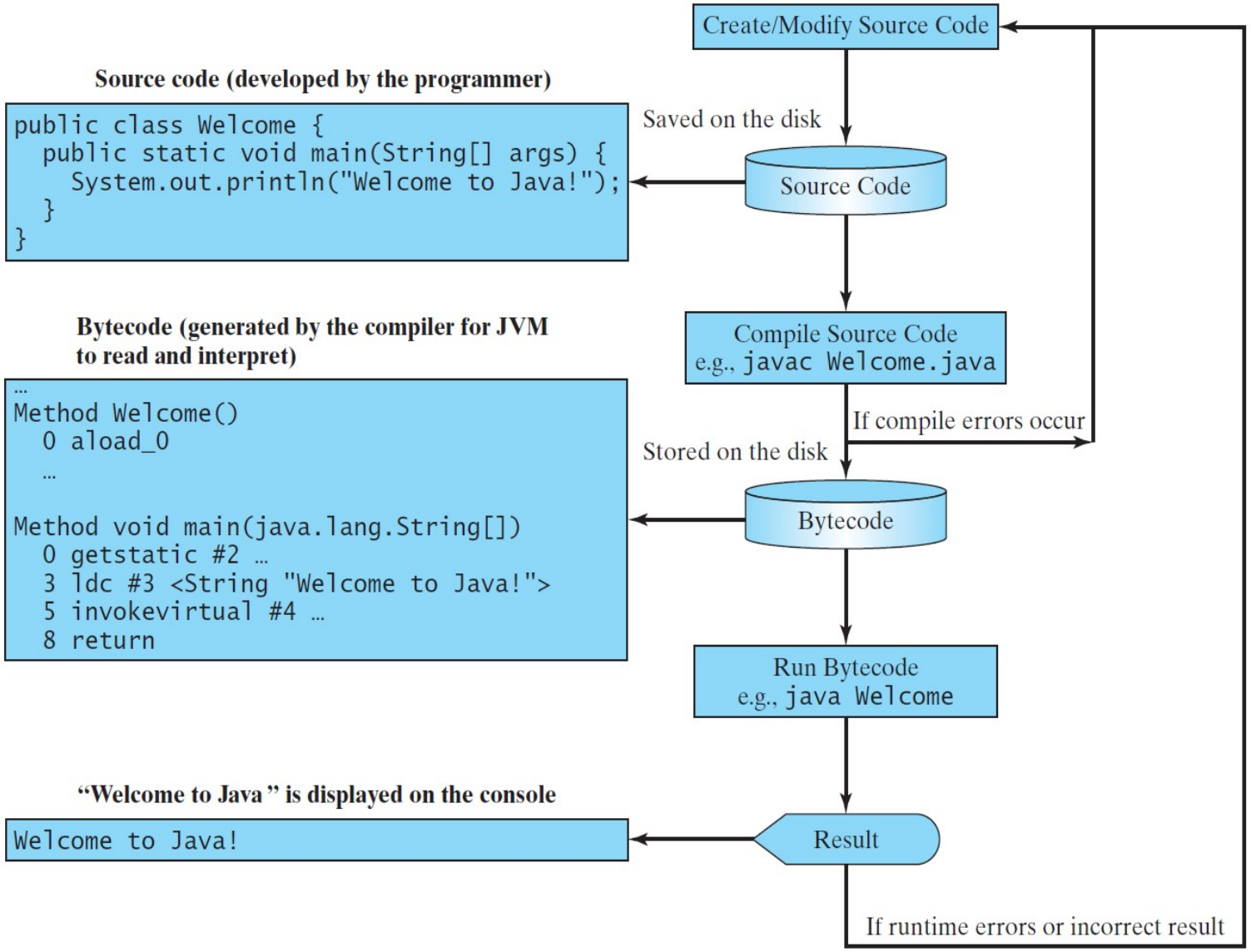


What is Java?

- Java is an object-oriented programming language.
- 1991 - Green Team started by Sun Microsystems.
- Designed to be for general purposes such as developing desktop applications, web applications, mobile applications, games etc.
- It also known as “Write Once Run Any Where” language (WORA).

Java program components

```java
public class Welcome {
    public static void main(String[] args) {
        // This program prints "Welcome to JAVA!"
        System.out.print("Welcome to JAVA!");
    }
}
```

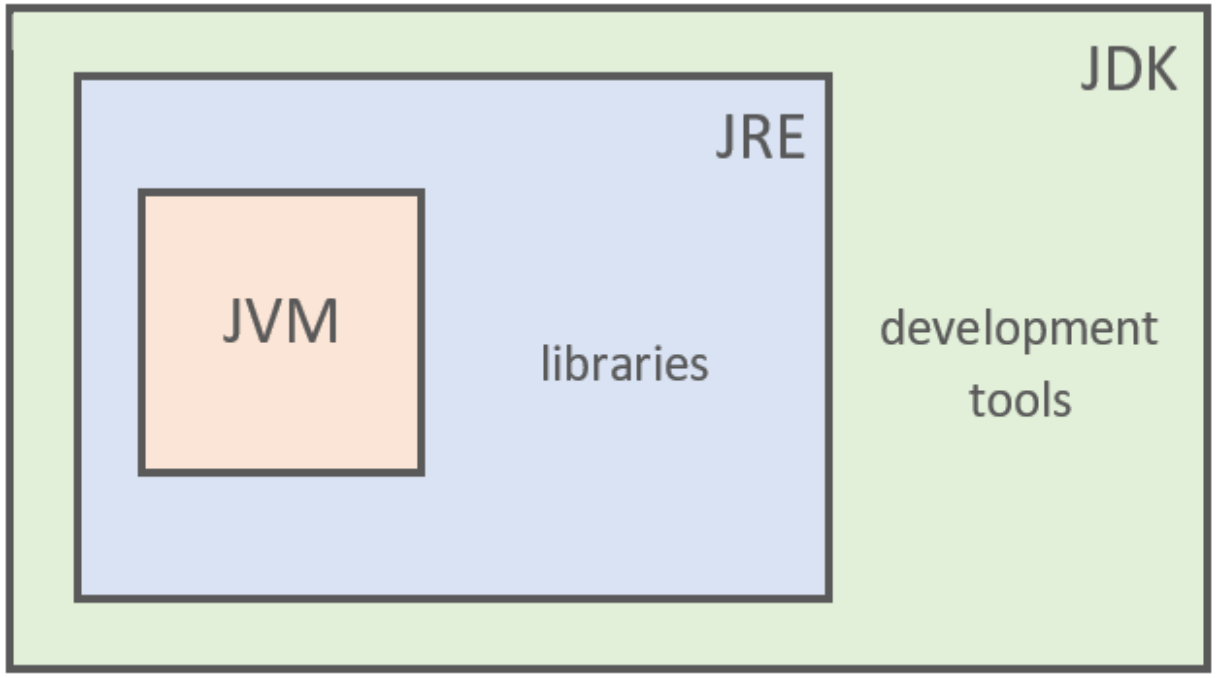


Java Development Editions

- Java Standard Edition (J2SE)
  - J2SE can be used to develop client-side standalone applications or applets.
- Java Enterprise Edition (J2EE)
  - J2EE can be used to develop server-side applications such as Java servlets, Java ServerPages, and Java ServerFaces.
- Java Micro Edition (J2ME).
  - J2ME can be used to develop applications for mobile devices such as cell phones.
- This course uses J2SE.

Components of Java Platform

- There are three interrelated components for Java platform:
  - JDK (Java Development Kit)
  - JRE (Java Runtime Environment)
  - JVM (Java Virtual Machine)



JDK = JRE + Development Tools
JRE = JVM + Library Classes

Java Development Kit (JDK)

- The JDK is used to develop Java applications.
- JDK internally contains JRE.
- JDK contains
  - Compiler (javac)
  - Core class library
    - classes.zip
    - rt.jar

Java Development Kit (JDK)

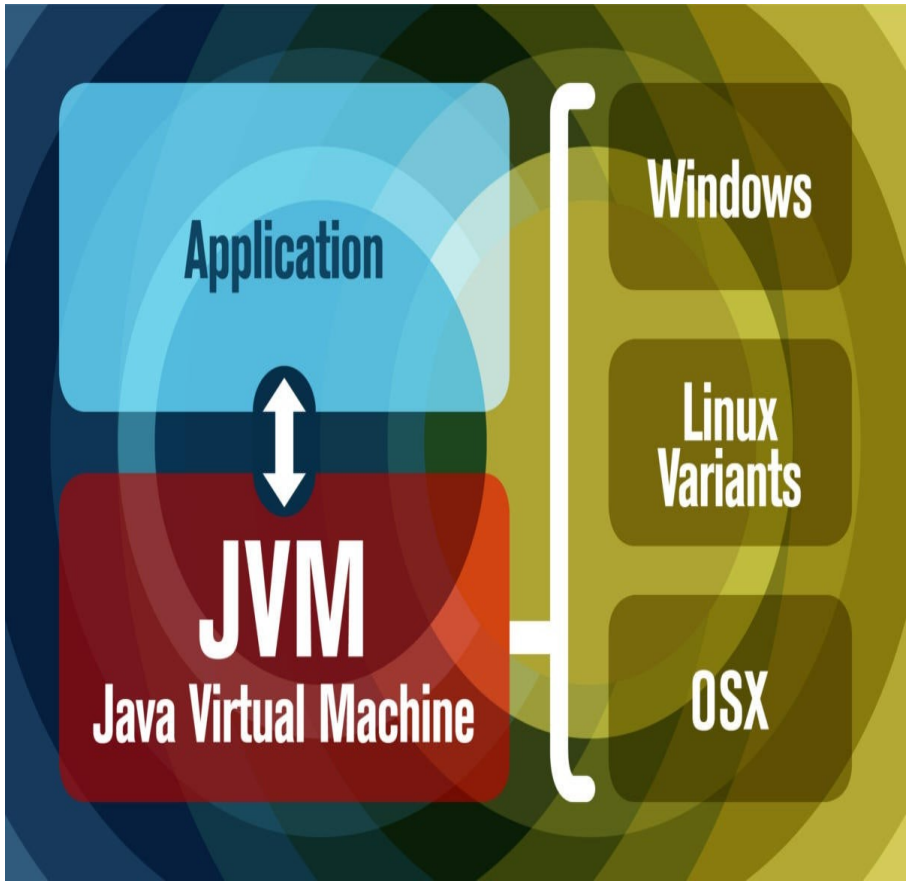
- JDK contains
  - Debugger (jdb)
  - Bytecode interpreter: The JVM (java)
  - Documentation generator (javadoc)
  - Java Archive utility (jar)
  - Others
- Without JDK we cannot build any Java program.
- Developers choose JDK by Java version and by package or edition.

Java Runtime Environment (JRE)

- The JRE is the responsible unit to run the Java program.
- Without JRE we cannot run Java program(JDK is used to build Java program, whereas JRE is used to run Java program).
- JRE contains JVM.
- JRE contains all the inbuilt packages and class library files(e.g. lang , io, util, etc).

Java Virtual Machine (JVM)

- JVM platform independent.
- JVM is available for multiple operating systems, and programs created with the JRE will run on all of them.
- JVM takes .class files and executes it by managing the memory.

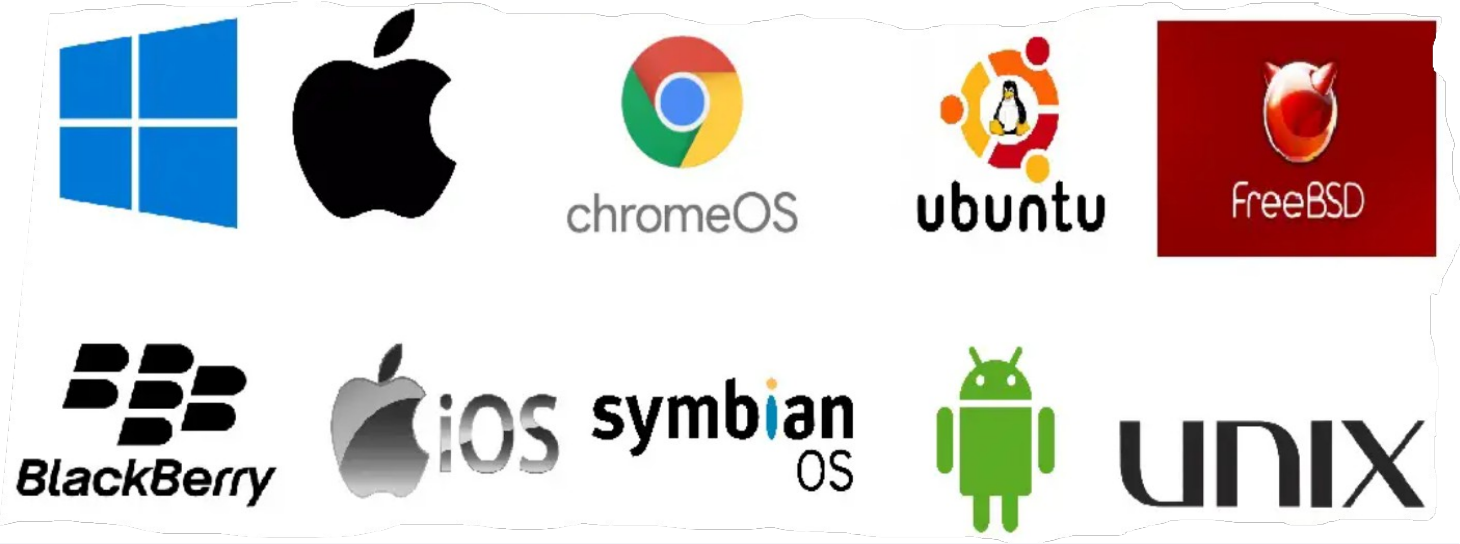


Java Virtual Machine (JVM)

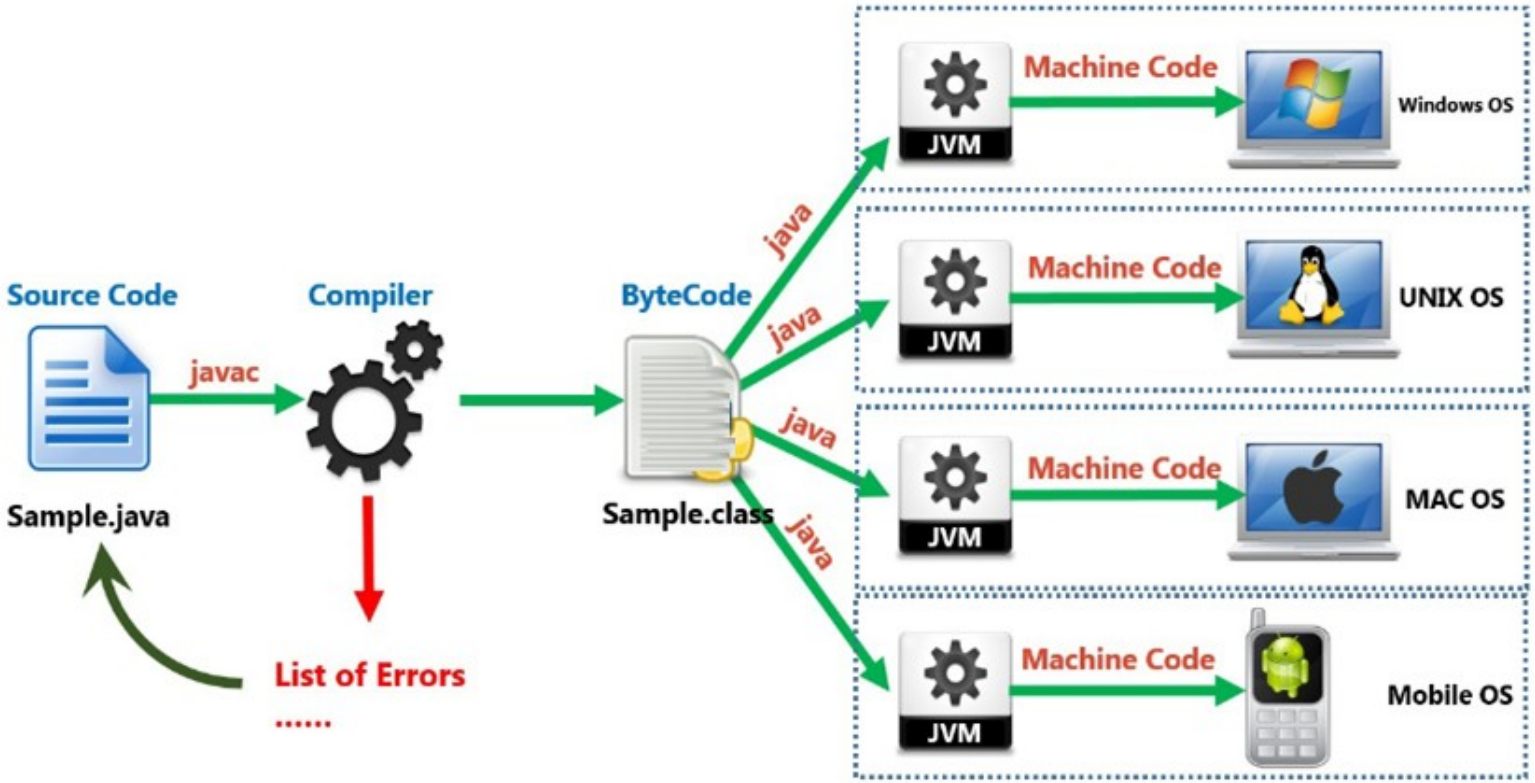
- JVM contains JIT (Just In Time Compiler), which improves the performance of Java applications at run time
- JVM loads, verifies, and executes the code and provides the runtime environment.
- JVM plays a major role in Java memory management.

Java As Independent Computing Platform (Portability)

- Portable means that a program may be written on one type of computer and then run on a wide variety of computers, with little or no modification.
- Java byte code runs on the JVM and not on any particular CPU; therefore, compiled Java programs are highly portable.
- JVMs exist on many platforms:



Java As Independent Computing Platform (Portability)



Java IDE

- An integrated development environment (IDE) is a software application that helps programmers develop software code efficiently. It increases developer productivity by combining capabilities such as software editing, building, testing, and packaging in an easy-to-use application.
- NetBeans
- Eclipse
- blueJ
- intellJ
- In this course we will use NetBeans IDE.

References

- Introduction to Java Programming, Comprehensive Version, Student Value Edition, Daniel Liang, 12th Edition, Pearson 2019
- lecture Notes: edugate.asu.edu.jo.
- lecture Videos: YouTube channel.
- How to install JDK & NetBeans IDE manual.